

eCH-1234 Template Quarto Document

September 3, 2026

Abstract Brief summary of the purpose of the document.

Table of contents

Note	1
1 Introduction	2
1.1 Status	3
1.2 Scope of application	3
1.3 We can have sub-headings	3
1.3.1 Also sub-sub-headings	3
2 Technical notes	4
3 Data Model	5
3.1 Genre	6
3.2 Invoice	6
3.3 Music album	6
3.4 Music Recording	7
3.5 Organisation	7
3.6 Person	7
3.7 Quantitative Value	8
4 Data Retrieval	8
5 Safety considerations	10
6 Disclaimer	11
7 Copyrights	12
8 Annex A - References	13
9 Annex B - Cooperation and Verification	14
10 Annex C - Abbreviations and Glossary	15
11 Annex D - Changes in comparison to previous version	16
12 Annex E - Table of figures	17
13 Annex F - Table of tables	18

Note

This document uses a gender-neutral formulation when referring to persons. This is based on the guidelines (German) of the Federal Chancellery. Depending on the situation, paired forms (citizens), gender-abstract forms (insured person), gender-neutral forms (insured person) or paraphrases with-out personal reference are used. The generic masculine (citizen) is not permitted. Full forms are used in continuous texts, i.e. in texts consisting of formulated sentences. Short forms can be used in abbreviated text passages, namely in tables. The short form is used with a slash but without an ellipsis (referent). Gender asterisks and similar spellings are not used.

1 Introduction

1.1 Status

Approved: This document was approved by the Experts' Committee. It has normative power for the defined field of application in the determined scope of application.

1.2 Scope of application

The information in this chapter should provide the reader with a brief overview of what this standard is intended for. Information about the following matters may be helpful here.

1.3 We can have sub-headings

And write some text.

1.3.1 Also sub-sub-headings

And write some more text. Maybe even with a pretty image.



Figure 1: Always add some text to describe what the image shows.

When displaying diagrams, try to write them in Mermaid JS straight away; this makes changes in the future or translations straightforward.

2 Technical notes

We use the ROBOT CLI in our project (Jackson et al. 2019), especially for it's ability to run the HermiT reasoner (Glimm et al. 2014).

3 Data Model

3.1 Genre

A musical category in the Chinook dataset.

Target Class: `:Genre`

Table 1: properties Genre

Description	Path	Type	Cardinality
Name	<code>schema:name</code>		1..1
Part of	<code>schema:partOf</code>	<code>:Genre</code> or <code>sh:IRI</code>	0..*

3.2 Invoice

A purchase receipt in the Chinook dataset.

Target Class: `schema:Invoice`

Table 2: properties Invoice

Description	Path	Type	Cardinality
Customer: An invoice must be linked to exactly one customer.	<code>schema:customer</code>	<code>schema:Person</code>	1..1
Total payment due: An invoice must define a total payment due as a <code>schema:QuantitativeValue</code> .	<code>schema:totalPaymentDue</code>	<code>schema:QuantitativeValue</code>	1..1
Has part: An invoice must have at least one line item (<code>OrderItem</code>).	<code>schema:hasPart</code>	<code>schema:OrderItem</code>	1..*
Date created	<code>schema:dateCreated</code>	<code>xsd:date</code>	0..*

3.3 Music album

A collection of tracks in the Chinook dataset.

Target Class: `schema:MusicAlbum`

Table 3: properties Music album

Description	Path	Type	Cardinality
Name: Every album must have a name.	<code>schema:name</code>	<code>xsd:string</code>	1..1
Artist: Person or music group who created the album.	<code>schema:byArtist</code>	<code>schema:Person</code> or <code>schema:MusicGroup</code>	0..*

3.4 Music Recording

A single music track in the Chinook dataset.

Target Class: `schema:MusicRecording`

Table 4: properties Music Recording

Description	Path	Type	Cardinality
Name: Every track must have a name.	<code>schema:name</code>	<code>xsd:string</code>	1..1
In Album: A track can only belong to a valid <code>schema:MusicAlbum</code> .	<code>schema:inAlbum</code>	<code>schema:MusicAlbum</code>	0..*
Author: The person or group who wrote the track.	<code>schema:author</code>		0..*
Genre	<code>schema:genre</code>	<code>:Genre</code>	1..1
Duration: Track duration must be expressed as a <code>schema:QuantitativeValue</code> .	<code>schema:duration</code>	<code>schema:QuantitativeValue</code>	0..*
Content Size	<code>schema:contentSize</code>	<code>schema:QuantitativeValue</code>	0..*

3.5 Organisation

A company or organization in the Chinook dataset.

Target Class: `schema:Organisation`

Table 5: properties Organisation

Description	Path	Type	Cardinality
Name: Every organisation must have a name.	<code>schema:name</code>	<code>xsd:string</code>	1..1

3.6 Person

Any person (employee, customer, or custom person) present in the dataset.

Target Class: `schema:Person`

Table 6: properties Person

Description	Path	Type	Cardinality
Given Name: Every person must have a given name.	<code>schema:givenName</code>	<code>xsd:string</code>	1..1
Family Name: Every person must have a family name.	<code>schema:familyName</code>	<code>xsd:string</code>	1..1
Email Address: If an email is provided, it must follow a standard email format.	<code>schema:email</code>	<code>xsd:string</code>	0..*
Birth date: A person should have a valid birth date.	<code>schema:birthDate</code>	<code>xsd:date</code>	0..*
Address	<code>schema:address</code>	<code>schema:PostalAddress</code>	0..*
Works for: An employee can report to another person.	<code>schema:worksFor</code>	<code>schema:Person</code> or <code>schema:Organization</code>	0..*
Job title	<code>schema:jobTitle</code>		0..1
knows	<code>schema:knows</code>	<code>schema:Person</code>	0..*

3.7 Quantitative Value

A numerical value with an associated unit.

Target Class: `schema:QuantitativeValue`

Table 7: properties Quantitative Value

Description	Path	Type	Cardinality
Value: A quantitative value must have exactly one numeric value.	<code>schema:value</code>		1..1
Unit Code: A quantitative value must specify its unit via a unitCode URI.	<code>schema:unitCode</code>	<code>sh:IRI</code>	1..1

4 Data Retrieval

The master and reference data underlying this document are available as *Linked Data*.

The technological basis for this is the **Resource Description Framework (RDF)**, a central standard of the World Wide Web Consortium (W3C) for modeling data structures on the web. In RDF, information is not represented in classic tables, but as interconnected graphs. Each statement consists of a so-called triple (subject, predicate, object). This structure enables a machine-readable, interoperable, and cross-system unambiguous description of resources and their relations to one another.

For the storage and publication of this RDF data, **LINDAS** (Linked Data Service) is used, the official Linked Data service of the Swiss Federal Administration. LINDAS functions as a so-called *triple store*, a specialized graph database optimized for the efficient storage and querying of RDF triples, which makes the data publicly available via a standardized interface.

The following chapter provides minimal instructions on how the data can be queried and retrieved from LINDAS.

```
BASE <https://example.org/>
PREFIX schema: <http://schema.org/>
SELECT *
WHERE {
    ?genre a <Genre> ;
    schema:name ?name .
}
LIMIT 10
```

The underlying data itself is maintained on GitHub as Turtle files.

```
@base <http://example.org/> .
@prefix genre: <http://example.org/genre/> .
@prefix schema: <http://schema.org/> .

genre:1 a <Genre> ;
    schema:name "Rock" .

genre:2 a <Genre> ;
    schema:name "Jazz" .
```

```
genre:3 a <Genre> ;  
  schema:name "Metal" ;  
  schema:partOf genre:1 .
```

5 Safety considerations

Information about the explicitly relevant legal bases or a note that during the implementation the relevant legal bases must be observed.

6 Disclaimer

eCH-standards which the registered association eCH provides the user free of charge or which make reference to eCH shall only have the status of recommendations. The registered association eCH will not be liable in any event for any decisions made or measures taken by the user based on these documents. The user will be responsible for verifying the documents himself prior to their use and to seek advice if required. eCH-standards can and shall not replace the technical, organizational or legal advice in the individual case.

Documents, procedures, methods, products and standards that are made reference to in eCH-standards are possibly protected by trademarks, copyrights or patents. It is the exclusive responsibility of the user to obtain the necessary licences from the entitled persons and/or organizations.

Although the registered association eCH has taken adequate care to prepare the eCH-standards with due diligence, it cannot grant any warranty or guarantee that the information and documents provided are up-to-date, complete, true or without any errors. eCH reserves the right to change the contents of the eCH-standards at any time and without prior announcement.

Any liability for damage caused by the use of the eCH-standards by the user shall be excluded to the extent legally admissible.

7 Copyrights

Persons preparing eCH-standards shall remain the owners of their intellectual property rights. These persons, however, obligate themselves to provide their intellectual property rights or other rights in third party intellectual property rights, to the extent possible, to the relevant technical units and the registered association eCH for free and for unlimited use and further development as part of the purpose of the association.

The standards prepared by the technical units can be used, distributed and developed further for free and to an unlimited extent by stating the name of the respective author of eCH.

eCH-standards are fully documented and free of any restrictions of licence and/or patent law. The associated documentation can be requested for free. These provisions shall apply to the standards prepared by eCH only, however, not to any standards or products of third parties which include reference to eCH-standards. The standards include the relevant references to third party rights.

8 Annex A - References

Glimm, Birte, Ian Horrocks, Boris Motik, Giorgos Stoilos, and Zhe Wang. 2014. "HermiT: An OWL 2 Reasoner." *Journal of Automated Reasoning* 53 (3): 245–69.

Jackson, Rebecca C, James P Balhoff, Eric Douglass, Nomi L Harris, Christopher J Mungall, and James A Overton. 2019. "ROBOT: A Tool for Automating Ontology Workflows." *BMC Bioinformatics* 20 (1): 407.

9 Annex B - Cooperation and Verification

10 Annex C - Abbreviations and Glossary

11 Annex D - Changes in comparison to previous version

12 Annex E - Table of figures

13 Annex F - Table of tables